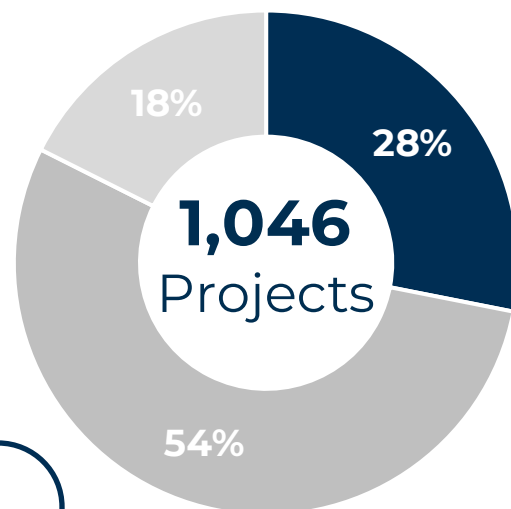




Internet of Value is the set of applications focused on value exchange, such as **cryptocurrencies**, **stablecoins** and virtual currencies promoted by central banks - **CBDCs**



- Internet of Value
- Blockchain for business
- Decentralized web

Methodology:



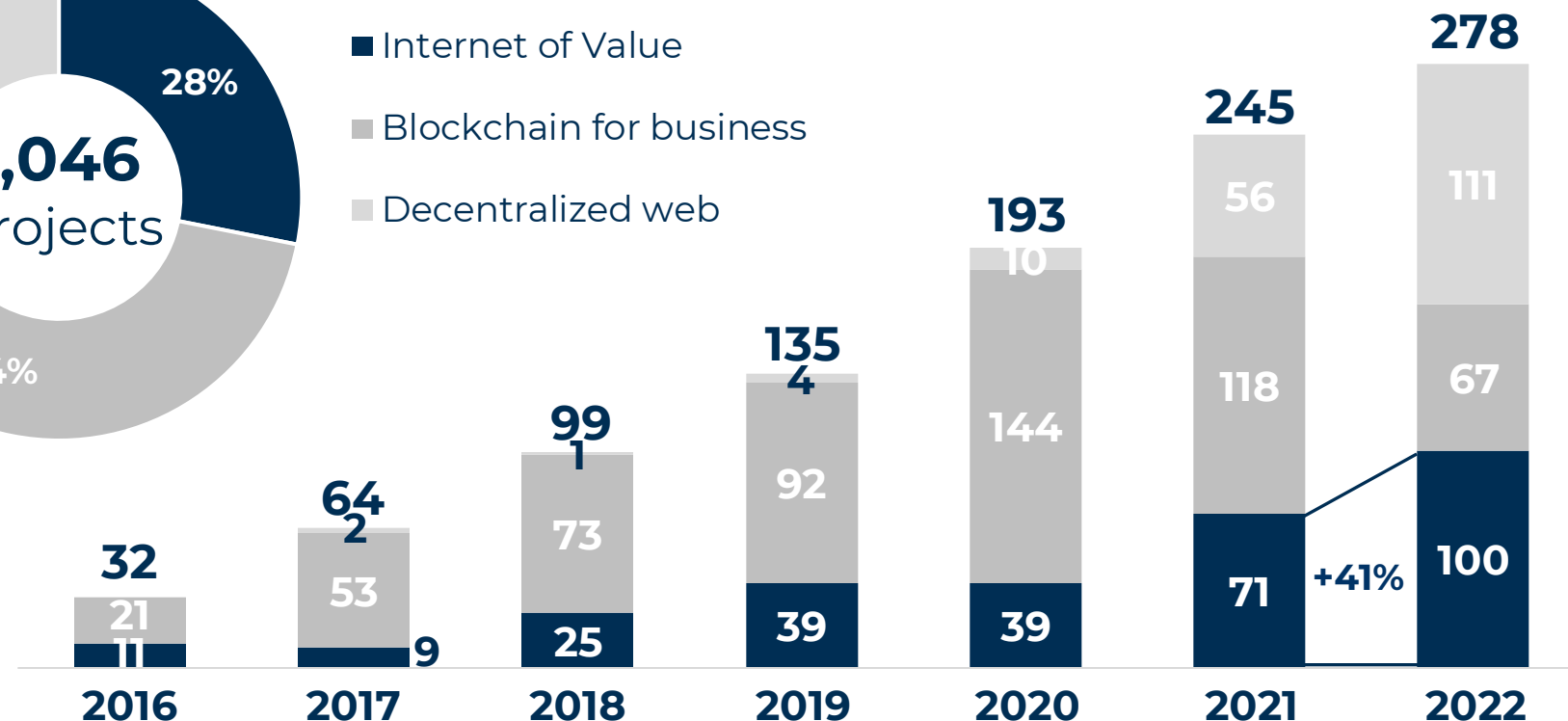
Borders: world

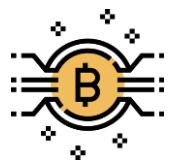


Criteria: Blockchain projects by companies and institutions

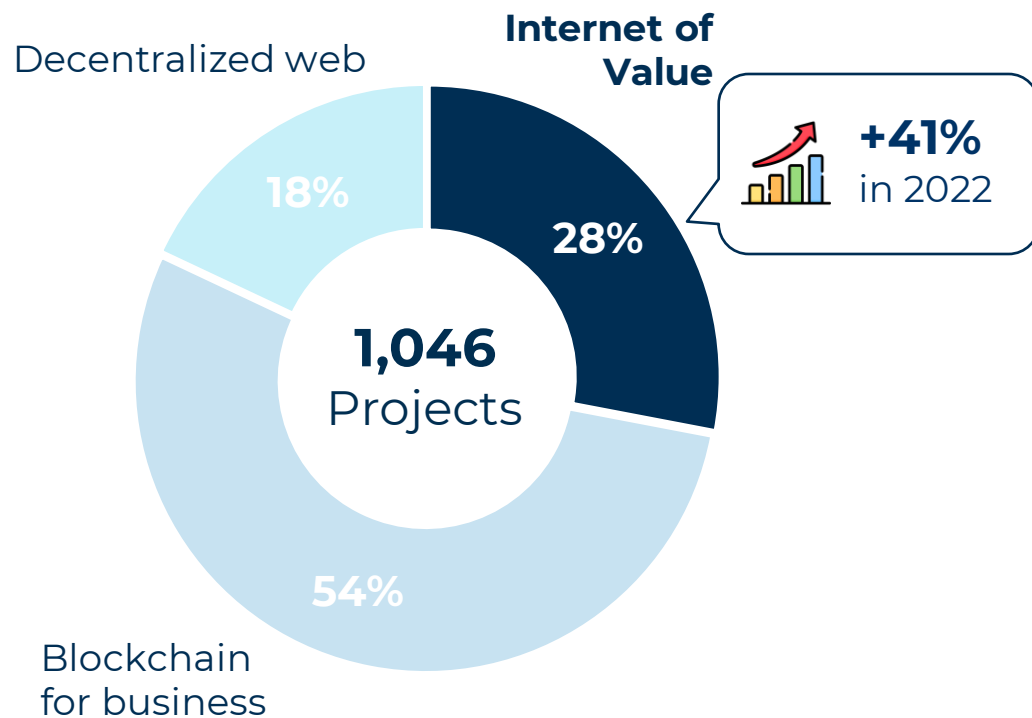


Sources: Generalist and industry-specific press





The **introduction of regulation (MiCAr)** and the **adoption by large industry players** can push the adoption further.



- Markets in Crypto-Assets regulation (MiCAr) approved



- Traditional actors begin to integrate **cryptocurrencies**

stripe



BlackRock

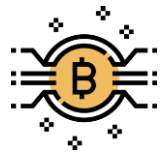
VISA

- Financial institutions start leaning towards **stablecoin** issuance

SOCIETE GENERALE

MUFG





The Internet of Value since 2009 has grown steadily, today (26/10/2023) cryptocurrencies and stablecoins alone reached **1,280\$ of capitalization**. Furthermore **94 central banks are working on CBDCs** projects (>59% of the world).

Cryptocurrencies and stablecoins

10%
of European
households own
crypto-assets⁽¹⁾

1,280 bln\$
Total Market
Capitalization⁽²⁾

125 bln\$
Total Stablecoins
Capitalization⁽²⁾

Central Bank Digital Currencies⁽³⁾



129 initiatives
(102 national and 27 international)



94 central banks
(~59% del totale)



63/100 banks
developed at least
one IoV project



CBDC

Projects related to the issuance, use, custody of Central Bank Digital Currencies



Stablecoins

Projects related to the issuance and use of stablecoins & deposit tokens



Cryptocurrencies

Projects related to investment and custodian services

Methodology:



Borders: world

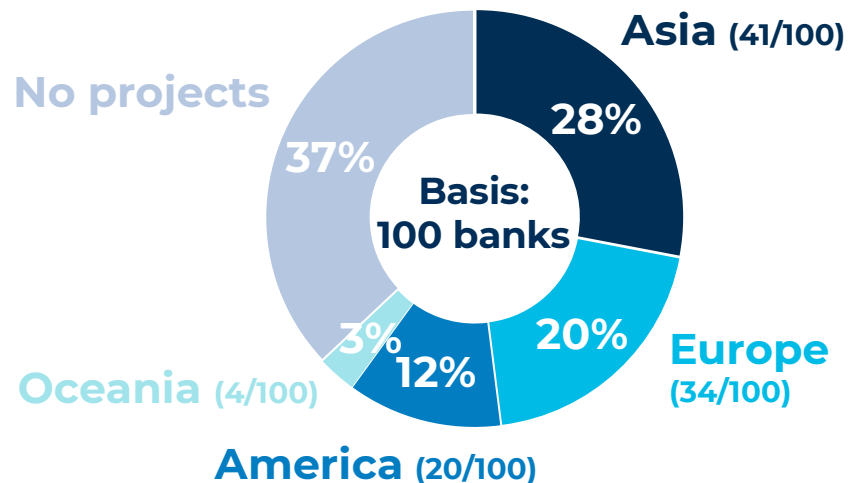


Criteria: Top 100 largest banks for AUM at 31/03/2023

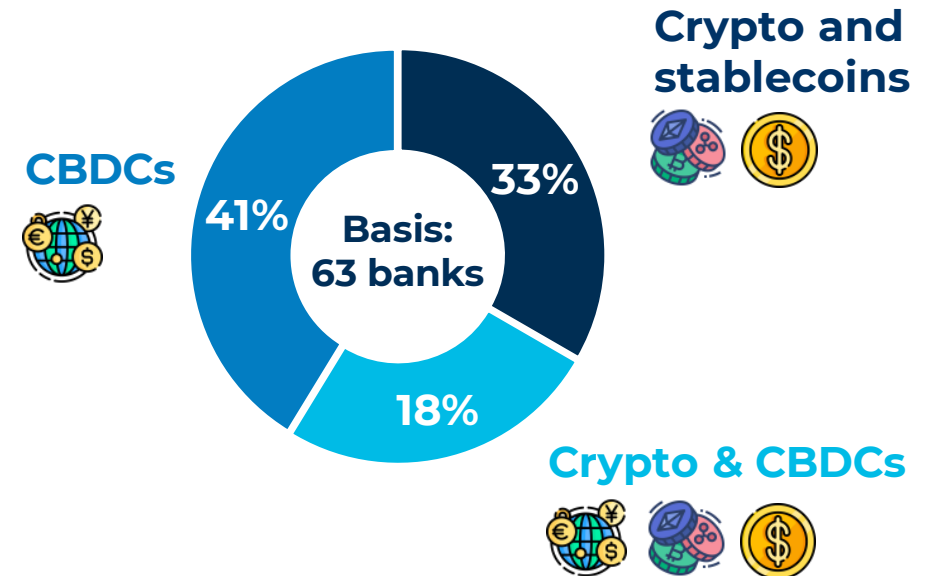


Sources: S&P Global, Generalist and industry-specific press

Where is IoV being adopted more?



What are banks focusing on?





Over the years, the Internet of Value (IoV) has seen an evolution. Nowadays **new assets** such as stablecoins and central bank digital currencies (CBDCs) together with the , as well as the **institutionalization of digital assets** represent the key trends within it.



Birth of Bitcoin



2009

J.P.Morgan



Companies and institutions

Traditional players start experimentations on the Internet of Value.

2017-2019



Policy & regulation

Countries around the world start developing regulation on digital assets.



2020-2022



CBDCs projects

Exploration of CBDCs projects all around the world. Launch of the firsts 2 real CBDCs.



2023



Institutional IoV on public ledgers

Surge of IoV projects by traditional actors on public ledgers.

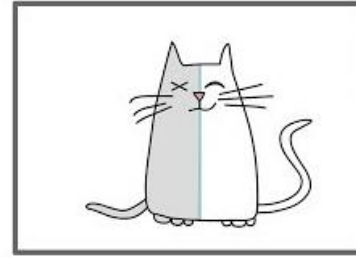


Media hype

Birth of new platforms and assets spreads the idea that Blockchain may be one of the technologies that will revolutionize digital.

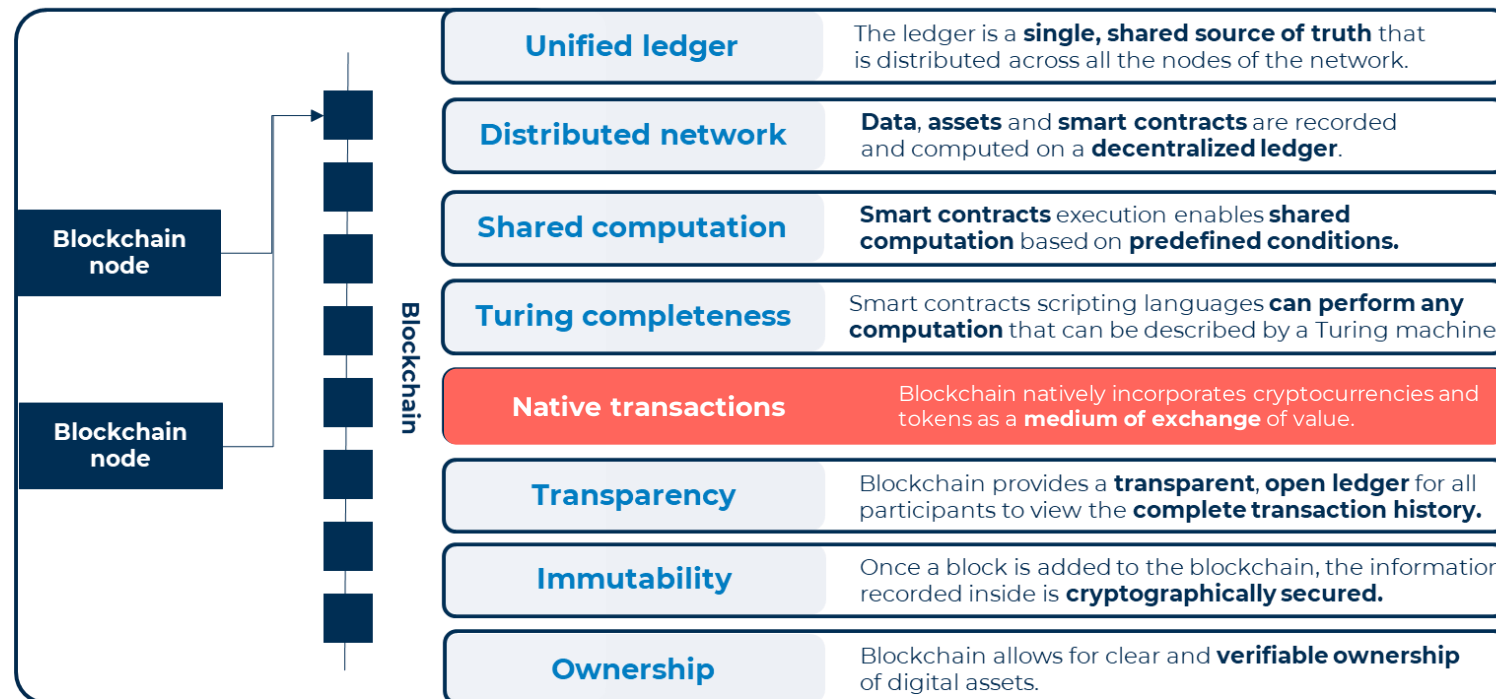


Schrödinger's Cat



It already exists and it is live....

...but at the same time it is not really available



Infrastructure Layer



European **regulators** and public
authorities can play
a **key role**

- **Support** European players (both *native* and *traditional* ones) in developing web3 innovations
- **Develop** the right regulatory frameworks crafted on the peculiarities of distributed technologies, without limiting innovation
- **Ensure** the existence of a legitimate, legally recognized form of **money to be used in web3**



Why the interest on Internet of Value from native e traditional players?

Osservatorio Blockchain & Web3

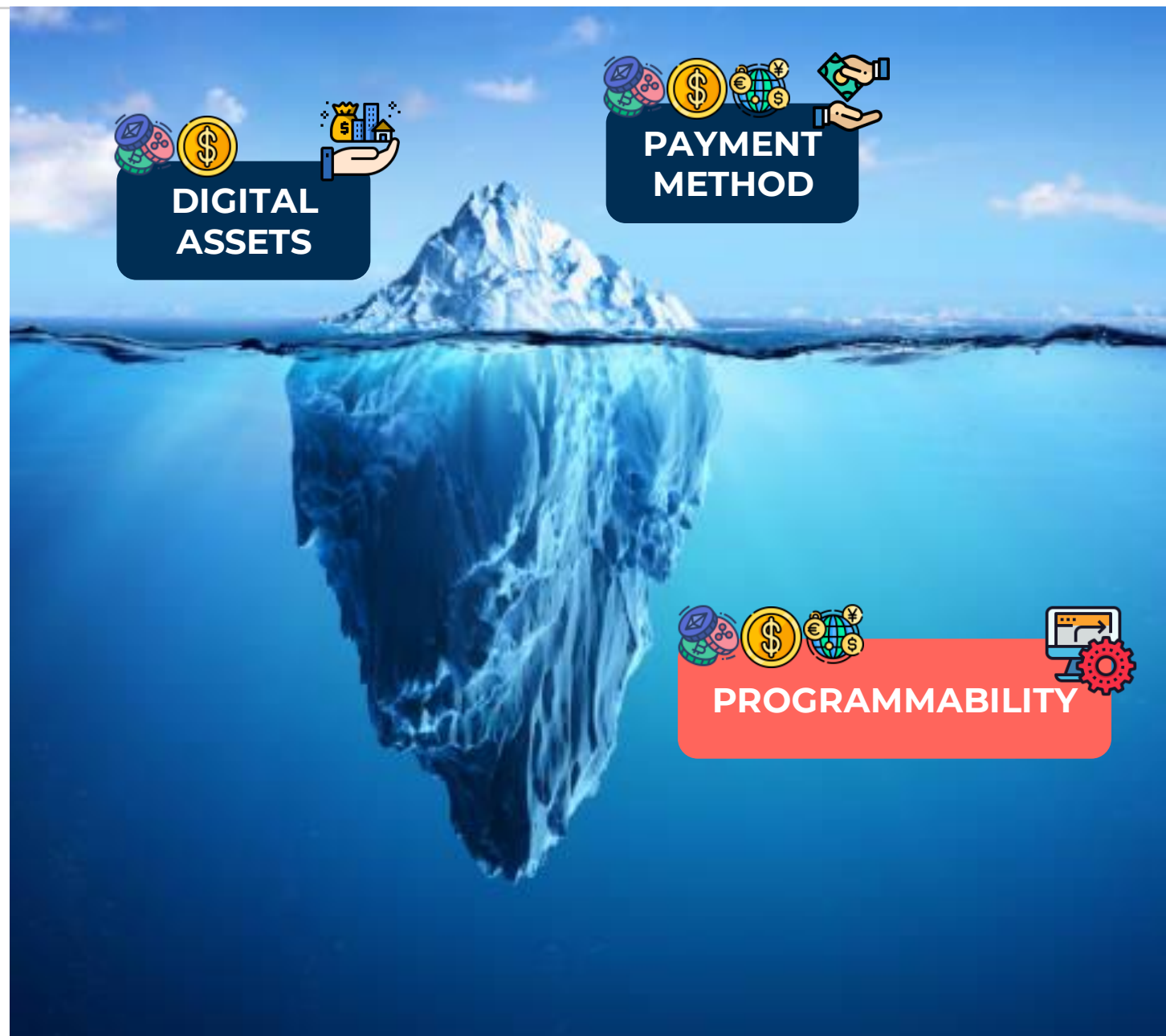
14.11.23



#OSSBLOCKCHAIN23



Is Internet of Value only a new way of storing and transferring value?



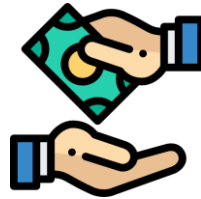
POLITECNICO
MILANO 1863

SCHOOL OF MANAGEMENT
DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA



osservatori.net
digital innovation

These terms are often used interchangeably,
although their meanings are different



Payments programmability

Ability to create and customize payment processes, enabling automation, conditions, and rules for payments.



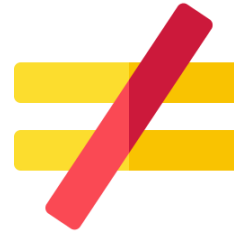
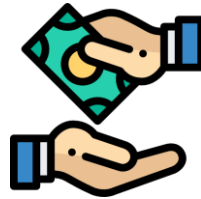
Currency programmability

Ability to customize digital currencies or tokens. Allows for the creation of specific features and rules for a particular currency.

These terms are often used interchangeably
although their meanings are different



The European Central Bank has publicly emphasized that the **Digital Euro will not include currency programmability.**



Payments programmability

Currency programmability



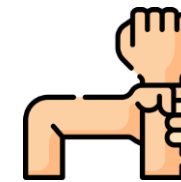
Direct debit



APIs



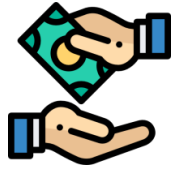
Smart contract



Expendability
conditions



Expiry
date



Payments
programmability



Automation



APIs



Smart contracts



Transparency

Code is private

Open-source code



Correct execution

Granted by third party

Granted by the network



Composability

Requires ad hoc development

Built-in within the protocol



Downtimes

Runs on cloud, or private servers

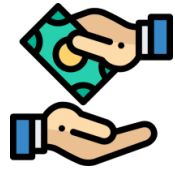
Runs on a distributed network



Payments

Requires licensed intermediary

Potentially implementable by anyone



Payments
programmability



Automation



APIs



Smart contracts



Transparency

Code is private



Correct execution

Granted by third party



Composability

Requires ad hoc development



Downtimes

Runs on cloud, or private servers



Payments

Requires licensed intermediary



Openness and transparency
facilitates a boost in innovation
through simpler introductions of
new ideas.

Built-in within the protocol

Runs on a distributed network

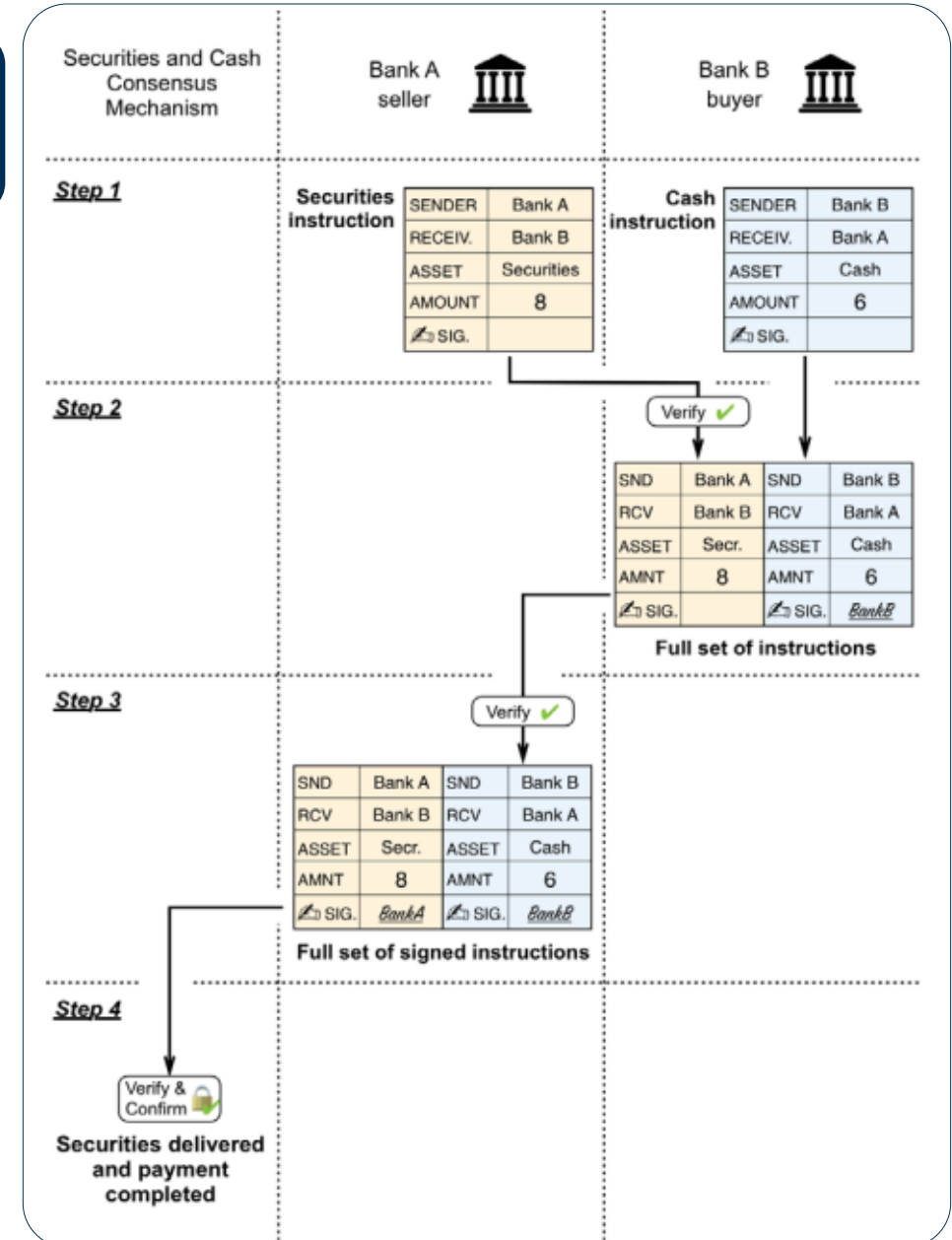
*Potentially implementable by
anyone*

What does programmability mean?



With **smart contract** payments programmability securities' exchange can be **automated and settled in seconds**.

What is delivery versus payment (DvP)?

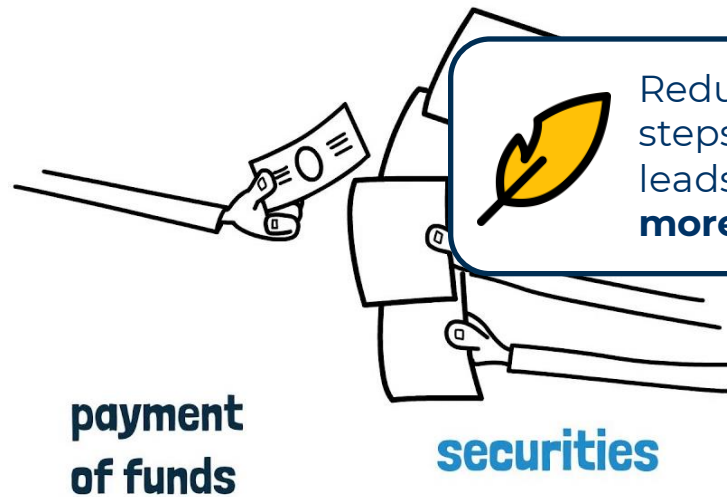


What does programmability mean?

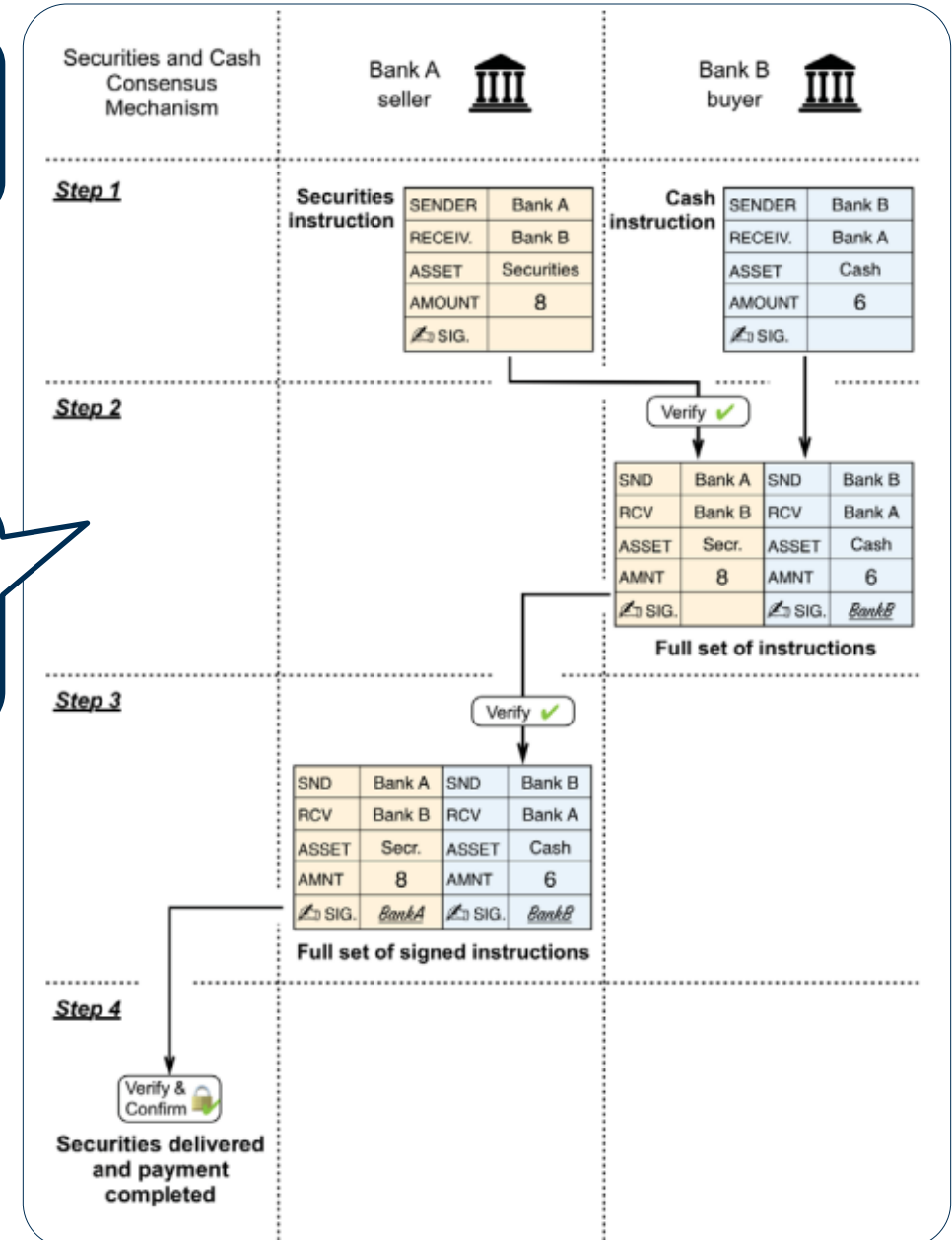


With **smart contract** payments programmability securities' exchange can be **automated and settled in seconds**.

What is delivery versus payment (DvP)?



Reducing the number of steps and actors involved leads to a **simpler and more efficient process**.





Dividends



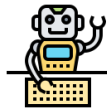
Gambling and competitions



Bond yields



Royalties



Machine-to-machine



Trading



Subscriptions



Trade finance



Rent



Insurance



Escrow accounts



Micro-payments



Lending



Dynamic pricing

There are **many different use cases**, most implications for business ecosystems, however, can't yet be seen.



Identity and **credentials**



Financial and real world **assets**



Reliable **cash-on-chain**

...



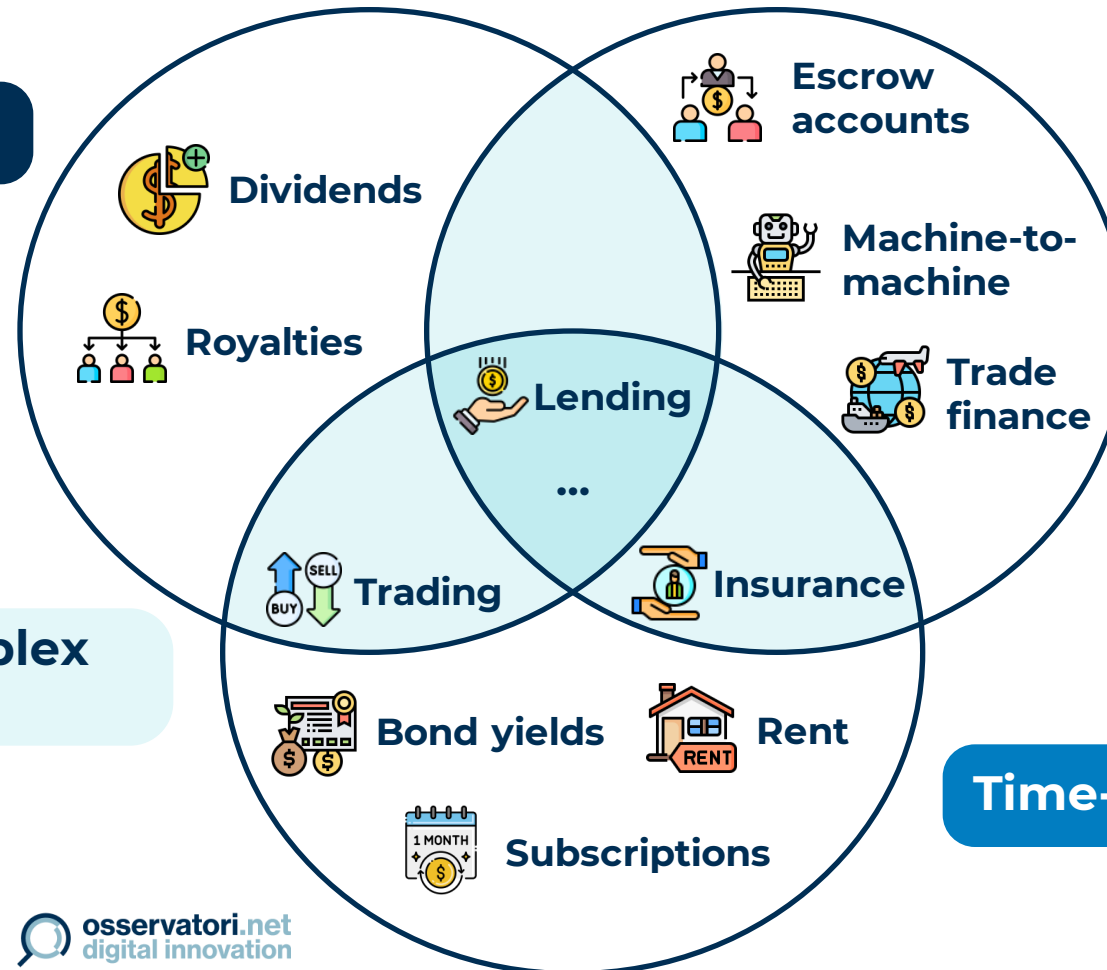
There are **different levels of complexity** that can be implemented within smart contracts. **Turing complete** programming languages allow for any logic to be implemented, **possibilities are therefore endless**.

Split payments

Delivery versus payment

More complex logic

Time-bound automation





There are **different levels of complexity** that can be implemented within smart contracts. **Turing complete** programming languages allow for any logic to be implemented, **possibilities are therefore endless**.

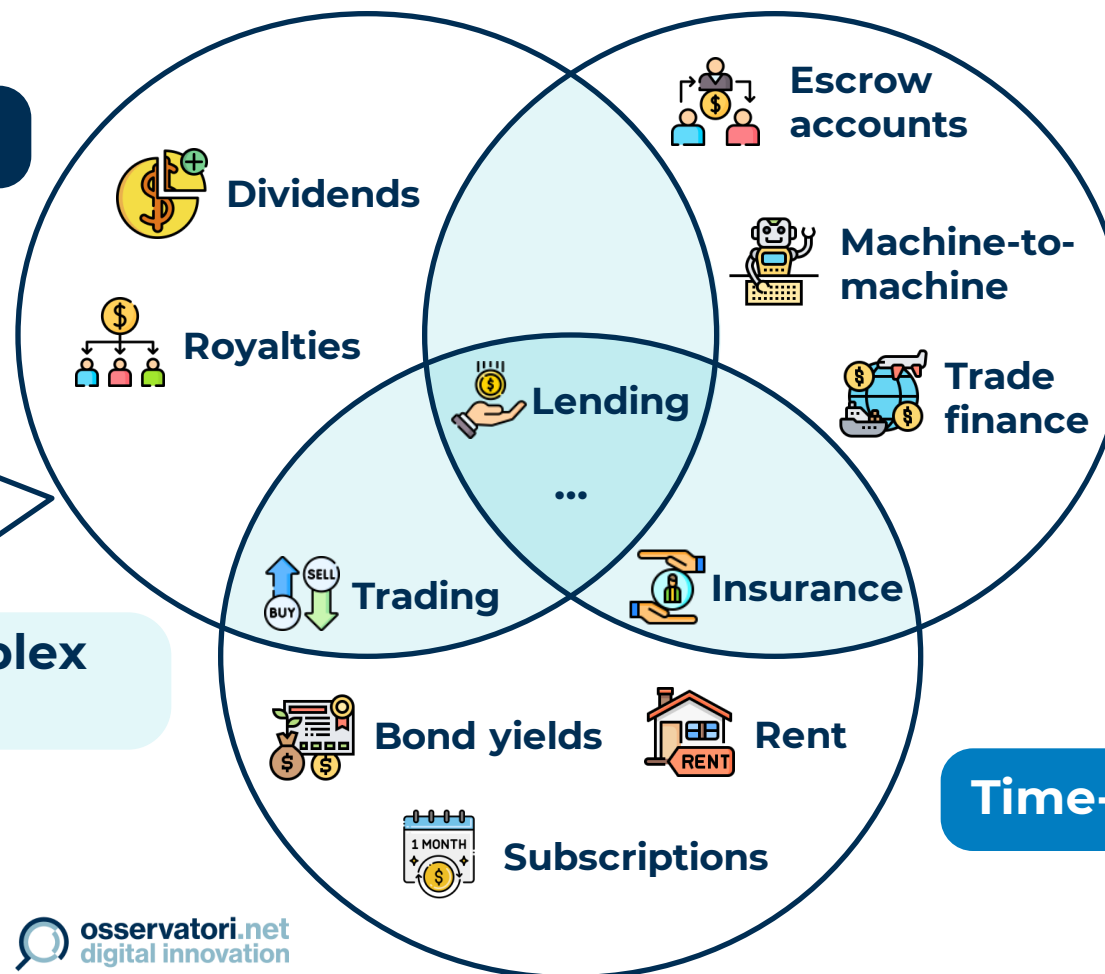
Split payments

Composability



Different processes and use cases can be combined among themselves.

More complex logic

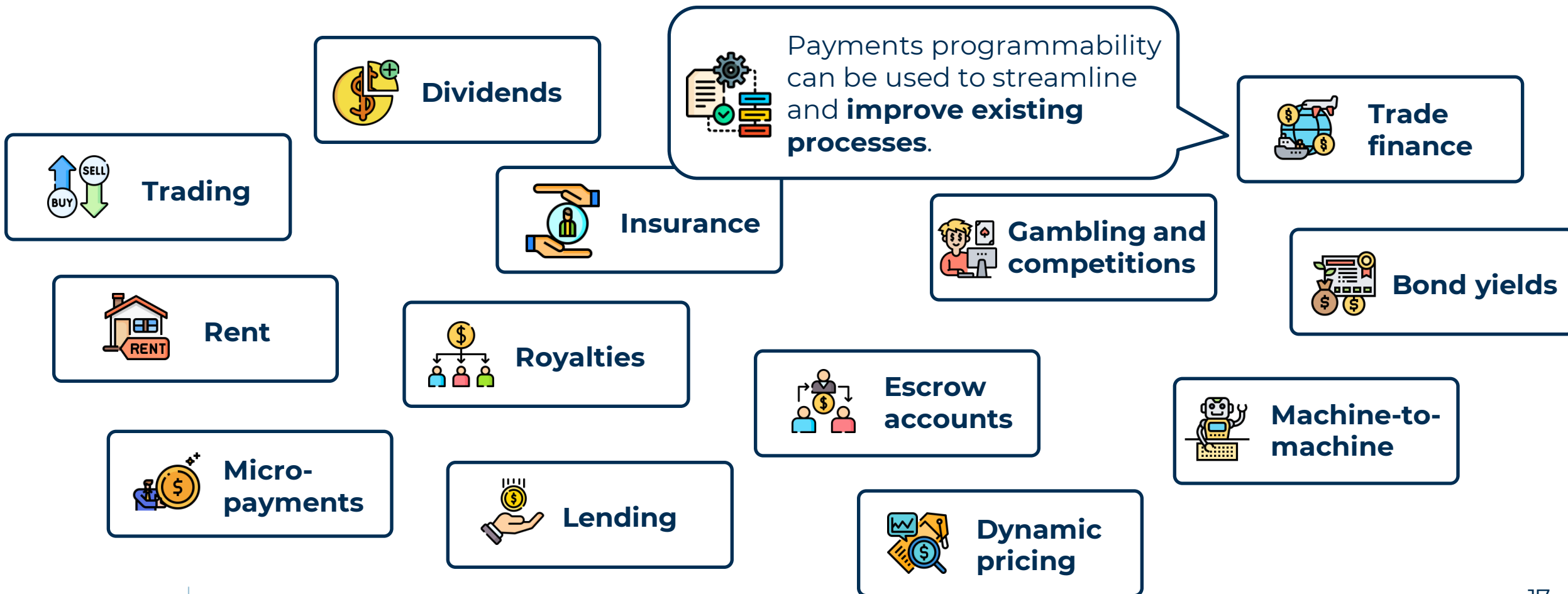


Delivery versus payment

Time-bound automation

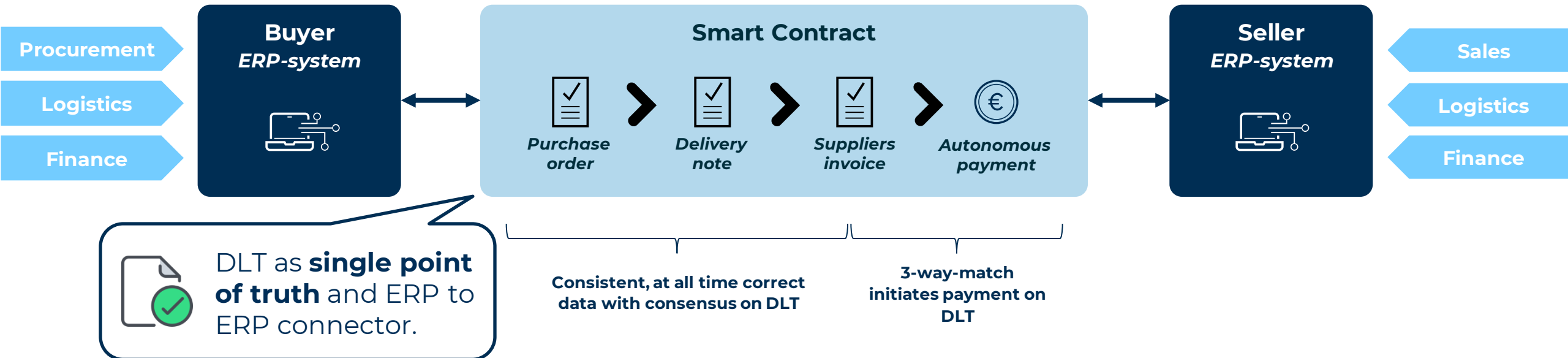


Transfer programmability use cases can streamline **existing processes** and enable **new business opportunities**





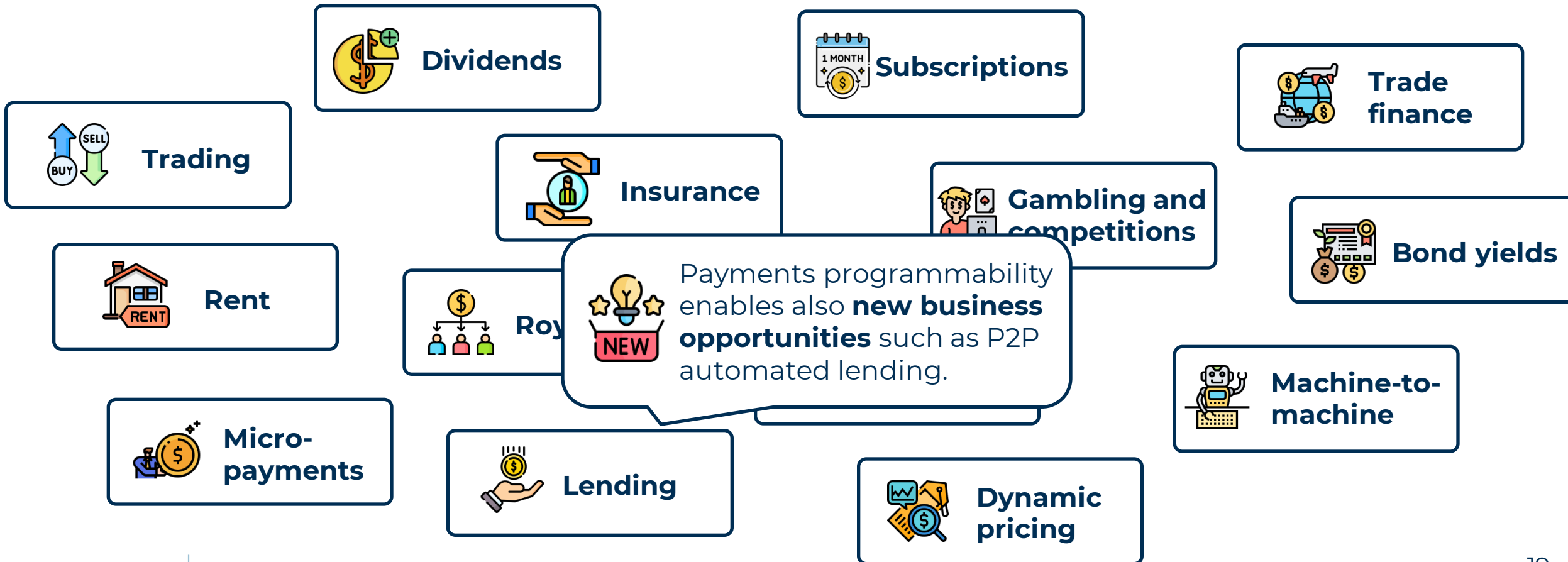
Themis has been designed to **assist supply chain interactions between buyer and seller.**



Process reliability and efficiency powered by tokenized DLT-based money

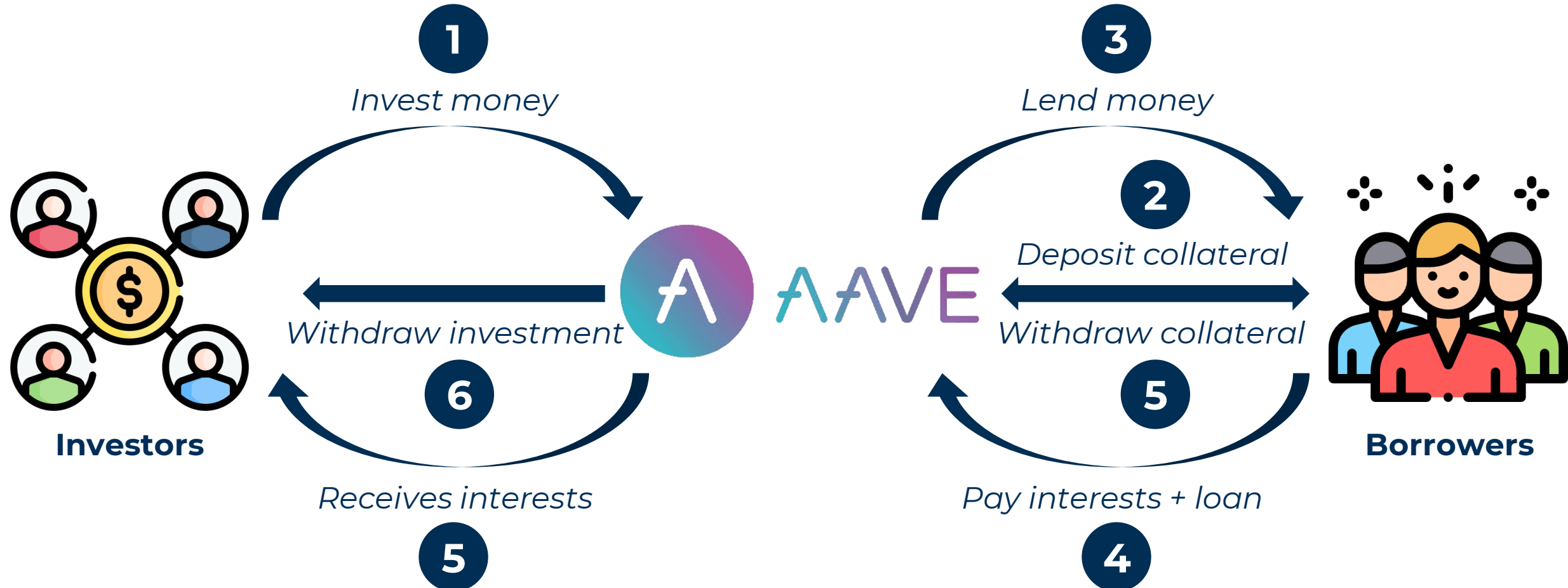


Transfer programmability use cases can streamline **existing processes** and enable **new business opportunities**





Automated lending platforms as Aave exploit payments programmability to enable **trustless peer-to-peer lending**.



	Split payments	Time-bound automation	Delivery versus payment	More complex logic
Energy				  *
Materials				
Industrials				   *
Consumer Discretionary	  		DAIMLER TRUCK *	
Consumer Staples				
Health Care				
Financials		  *	  a SIX company	  *
Information Technology				
Communication services				
Utilities				
Real Estate				

	Split payments	Time-bound automation	Delivery versus payment	More complex logic
Energy			   	 ngn engena *
Materials				
Industrials				 XDC TRADE NETWORK
Consumer Discretionary	 			 
Consumer Staples			 	
Health Care				
Financials	  	 	   	  
Information Technology				
Communication services	 	  	  OpenSea	 
Utilities				
Real Estate	 		 	

These opportunities need payments to **be programmable by anyone**. Today, **only instruments on public blockchains** satisfy this requirement.



*How will
programmability
be implemented?*

wCBDC



Stablecoin



Tokenized deposits



Crypto

